

1. (Exercise 1.1 from Tuckman & Serrat) The value of each coupon payment is given by

$$1,000 \left(\frac{.0275}{2} \right) = 13.75.$$

Let us label May 31, 2010 to be $t = 0$. Then the maturity of the bond is $T = 7$. The cash flows are given by

$$\$13.75 \text{ at } t = .5, 1, 1.5, \dots, 6.5 \quad \text{and} \quad \$1,013.75 \text{ at } t = 7.$$

2. (Exercise 1.5 from Tuckman & Serrat) We can replicate 100 face of the $3\frac{1}{2}$ s of 5/15/2020 by purchasing

$$100 \left(\frac{3.5}{8.75} \right) = 40$$

in face of the $8\frac{3}{4}$ s of 5/15/2020 plus 60 face of the 0s of 5/15/2020. The cost of the replicating portfolio is

$$\left(\frac{40}{100} \right) 145.67 + \left(\frac{60}{100} \right) 69.21 = 99.794.$$

The (arbitrage-free) price per 100 face of the $3\frac{1}{2}$ s of 5/15/2020 is therefore given by 99.794.

The on-the-run 10-year bond might sell for more than the arbitrage-free price in practice because of its superior liquidity and financing characteristics relative to the other bonds.

3. (Exercise 2.1 from Tuckman & Serrat) At the end of two years you will have

$$100 \left(1 + \frac{.02}{2} \right)^4 = 100(1.01)^4 = 104.06.$$

4. (Exercise 2.2 from Tuckman & Serrat) Let r denote the desired return. To find r , we solve the equation

$$107 = 100 \left(1 + \frac{r}{2} \right)^6,$$

which gives

$$r = 2 \left[\left(\frac{107}{100} \right)^{\frac{1}{6}} - 1 \right] = .022681.$$

5. For each positive integer i we have

$$d(i) = \frac{1}{(1.03)^{2i}} = \frac{1}{(1.0609)^i}.$$

Let us put

$$\lambda = \frac{1}{1.0609}.$$

The time-0 value of the prize is given by

$$25,000 \sum_{i=1}^{40} d(i) = 25,000 \sum_{i=1}^{40} \lambda^i.$$

Recall that (when $\lambda \neq 1$)

$$\sum_{i=1}^N \lambda^i = \frac{\lambda - \lambda^{N+1}}{1 - \lambda}.$$

Assuming that the first payment is made one year from today, we see that the time-0 value of the prize is

$$25,000 \left[\frac{\lambda - \lambda^{41}}{1 - \lambda} \right] = 371,930.58.$$

(We will review and derive the relevant formulas for summing geometric series in lecture.) If the first payment is made today, the time-0 value of the prize is 394,581.16.

6. (a) If the simple interest convention is used, the account balance will be

$$15,000(1 + (.06)23) = \$35,700.$$

- (b) If the interest is compounded annually, the account balance will be

$$15,000(1.06)^{23} = \$57,296.24.$$

- (c) If the interest is compounded quarterly, the account balance will be

$$15,000 \left(1 + \frac{.06}{4} \right)^{92} = \$59,015.64.$$

- (d) For monthly compounding, the account balance will be

$$15,000 \left(1 + \frac{.06}{12} \right)^{276} = \$59,418.86.$$

- (e) Under continuous compounding, the account balance will be

$$15,000e^{23(.06)} = \$59,623.53.$$

7. To find r we must solve the equation

$$1,000e^r + 2,000e^{.5r} = 3,128.98.$$

Let us put $x = e^{.5r}$ so that the equation becomes

$$1,000x^2 + 2,000x - 3,128.98 = 0.$$

Solving for x , we find that

$$x = 1.031989,$$

which gives

$$r = 2 \ln(1.031989) = .062976.$$

8. We have $3,637.44 = 5,000d(7)$, so that

$$d(7) = \frac{3,637.44}{5,000}.$$

It follows that

$$\hat{r}(7) = 2 \left[\left(\frac{5,000}{3,637.44} \right)^{\frac{1}{14}} - 1 \right] = .0459715.$$

$$\hat{R}(7) = \left(\frac{5,000}{3,637.44} \right)^{\frac{1}{7}} - 1 = .04649984.$$

9. The semiannual coupon payment for the bond is $.05(1,000) = 50$. Consequently we can replicate a zero with maturity $T = 5$ and face value \$20,000 by purchasing 20 coupon bonds and selling one annuity; the time 0 cost of the replicating portfolio is

$$20(1,100) - 8,000 = 14,000.$$

We conclude that

$$14,000 = d(5)20,000, \quad \text{i.e. } d(5) = .7,$$

which tells us that

$$\hat{r}(5) = 2 \left[\left(\frac{1}{.7} \right)^{\frac{1}{10}} - 1 \right] = .072622,$$

$$\hat{R}(5) = \left(\frac{1}{.7} \right)^{\frac{1}{5}} - 1 = .073941.$$